

(c) Find the following angles, stating all the angle properties.

(i) $\angle EBD$

(ii) $\angle DEA$

Answer (i) _____^o
(ii) _____^o [3]

(d) Explain why G is the center of the circle.

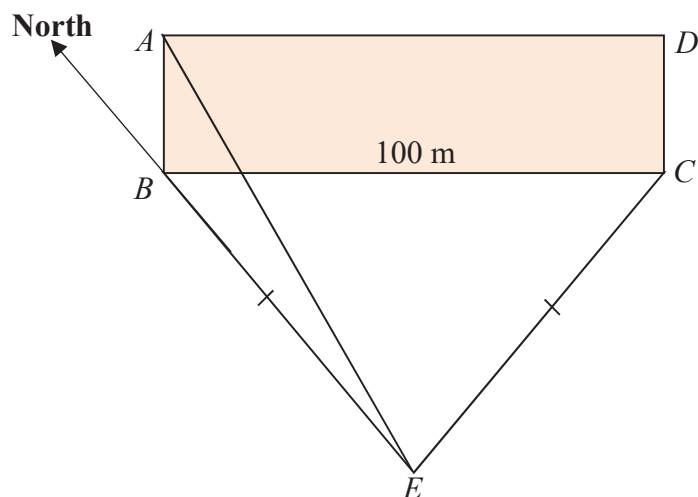
Answer

[2]

(e) If the radius of the circle is 7 cm, find the area of the shaded region.

Answer _____ cm^2 [3]

- 6 In the diagram, $ABCD$ represents a vertical cliff face. The bottom of the cliff, BC , is 100 m and is at sea level. A boat is in the sea at E .
 B is due north of E and the bearing of C from E is 070° .
 E is known to be equidistant from B and C .



- (a) Find the bearing of C from B .

Answer _____ $^\circ$ [2]

- (b) Show that $BE = 87.17$ m.

Answer

[2]

- (c) Find the area of $\triangle BCE$, leaving your answer to the nearest square metre.

Answer _____ m^2 [2]

- (d) It is given that the angle of depression of E from A is 20.2° . Find the height of the cliff BA , leaving your answers in 3 significant figures.

Answer _____ m [2]

- (e) P is a point on BC such that $\frac{\text{Area of } \triangle BEP}{\text{Area of } \triangle CEP} = \frac{3}{2}$.

Find the length of CP .

Answer _____ m [2]